

CHINA DATA CENTERS

The rise of Ulanqab as one of APAC's largest and fastest growing AI computing clusters

Recent news headlines on GW-scale data center investments by leading data center operators and internet/AI companies (e.g. GDS, DeepSeek, Z.AI, RedNote) have drawn investors' attention to **China's data center capacity expansion** outlook, and particularly to **Ulanqab**, a city that frequently comes up in news reports. In this note, we zoom in on Ulanqab, a prefecture-level city in Inner Mongolia that has become one of the largest and fastest growing computing clusters in China and APAC.

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Key findings

- 1. Data center capacity:** Ulanqab has received **12.5GW** capacity commitment from data center operators and internet/AI companies as of Jun 2026 (live and pipeline combined, [Exhibit 1](#)), implying **10x+** capacity expansion from its c.1.2GW live capacity as of 2025 which accounted for only c.4% of live capacity in China. This is **three times the total upcoming data center capacity in Johor, Malaysia** (live and pipeline combined, [Exhibit 2](#)) which has the largest incoming data center capacity in APAC (ex. China), according to Cushman & Wakefield.
- 2. Key players:** We see **GW-scale projects** underway from **3 of the top 4 data center operators in China** by live capacity, namely GDS, VNET and Chindata (owned by Dongyangguang, 600673.SS, Not Covered), as well as Envision (Private), followed by hundred MW-scale projects from ZDATA and Centrin (both Private). Meanwhile, in addition to **self-built campuses** owned by Kuaishou and UCloud, ByteDance, Z.AI and other leading internet/AI companies are reportedly aiming to establish a data center presence in Ulanqab.
- 3. Live demand:** The city's utilized capacity grew **70%+ yoy** to **c.800MW** as of end-2025 ([Exhibit 8](#)), while available computing power **more than doubled yoy** to **165 EFLOPS** as of Jun 2026 ([Exhibit 9](#)), equivalent to 5%/7% of China's utilized data center capacity/intelligent computing power, respectively. While the blended utilization rate (**66%** as of 2025) may not be representative considering rapid supply increase, we see VNET as one of the early movers in Ulanqab (c.300MW utilized capacity as of 1Q26 or 1/3 market share per GSe vs. 398MW live capacity, 388MW under construction and c.1GW held for future development) as an example to demonstrate robust demand; with 100% commitment and pre-commitment rates, its **average utilization rate improved**

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to 40%/80% within 2/4 quarters of capacity delivery ([Exhibit 11](#)).

4. **Electricity and water consumption:** Data center electricity power consumption grew **90%** yoy to 3.3TWh YTD as of Jun, equivalent to **7%** of the city's total power consumption (vs. 4.3% a year ago or 1.6% for China overall in 2025). While Ulanqab enjoys **rich green wind/solar power resources** (e.g. 1/10 of wind power in China) and favorable climate (e.g. average temperature 4.3°C), its **tight water resources** may have impacted the evolution of cooling systems, e.g. direct-to-chip liquid cooling.
5. **Green power:** While green power is not necessarily cheaper than regular grid electricity, data centers leverage green power to meet policy compliance requirements, including **green power mix** and **PUE (Power Usage Effectiveness) requirements**. **Direct green power supply** and **green power trading** are two primary ways to obtain green power. We see the following factors weighing on the economics of green electricity for data centers: i) generation/PPA prices; ii) green-certificate premium; iii) grid delivery & system charges; iv) dedicated-line and storage capex (direct connection only); and v) any extra backup power needed.

Implications for China's data center industry

1. **From “IDC pure-plays” to “IDC operations + green power asset management”:** Under the national policy of “coordination between computing power and electricity power”, we see more data center operators taking on roles as **green power asset managers**, i.e. optimizing renewable energy procurement, managing on-site generation, and hedging against power market volatility to secure low-cost, sustainable electricity. This echoes **VNET's “Hyperscale 2.0” vision** shared at its investor day in 2025 where data centers provide a platform-based, self-load-driven new power system vs. a project-based resource resale business in Hyperscale 1.0. Core competencies in Hyperscale 2.0 include: 1) direct green power connections and active microgrids; 2) large-scale energy integration to optimize energy utilization and grid stability; 3) smart power dispatching for both internal usage and external power spot trading; and 4) hardcore technology joint innovation focused on areas like power semiconductors, energy routers, and Solid State Transformer (SST) networking.
2. **From “cabinet leasing” to “electricity-to-token conversion”:** Service providers are moving up the value chain by adopting **token revenue-sharing** models, which was first publicized by GPU rental service providers such as Xingyun Technology (300209.SZ, Not Covered). On Jul 30, the company announced a supplementary agreement with one of its customers that includes: i) Rmb3.1bn contract value for 256 servers (18PFLOPS each, FP16) for 5 years, equivalent to a Rmb200k monthly rental rate; and ii) monthly fixed token service fees.
3. **Data center rental price increase likely capped by surging supply:** The 12.5GW total data center pipeline capacity in Ulanqab alone is equivalent to 45% of China's c.28GW live supply as of 2025. As mentioned in our prior report [Scaling the AI Infrastructure](#), we see that Inner Mongolia (including Ulanqab), Ningxia and Gansu have emerged as the fastest-growing hubs in terms of intelligent computing power to meet AI demand, which enjoy relative advantages in latency, electricity costs and

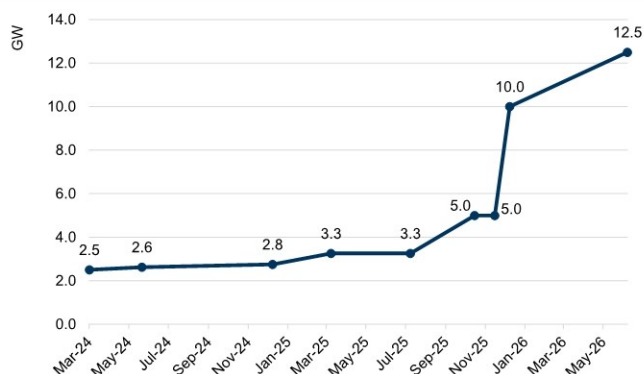
PUE. While we are bullish about China's AI demand and capex outlook, the data center supply advantage as part of China's Six Networks infrastructure program may imply lower likelihood of notable data center rental price increase.

- 4. Third-party data centers remain critical stakeholders to capture AI demand:** In Ulanqab, third-party data centers account for the majority of pipeline capacity and tend to claim bigger data center campuses vs. self-built facilities by internet/AI companies. In other words, we believe leading third-party data centers are well-positioned to capture rising AI demand thanks to their capex and power efficiency, in addition to vast capacity reserve in core computing hubs in China.

Data center capacity

Exhibit 1: Ulanqab has seen a surge in capacity commitment from 3.3GW as of Jul 2025 to 12.5GW as of Jun 2026...

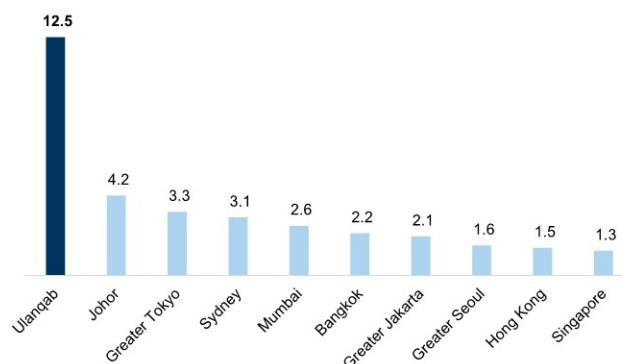
Total data center capacity commitment in Ulanqab



Source: Sina News, Local government websites, Data compiled by Goldman Sachs Global Investment Research

Exhibit 2: ...which is three times the total upcoming data center capacity in Johor

Total data center capacity (live, under construction and pipeline) in key APAC hubs (GW)



Source: Local government websites, Cushman & Wakefield

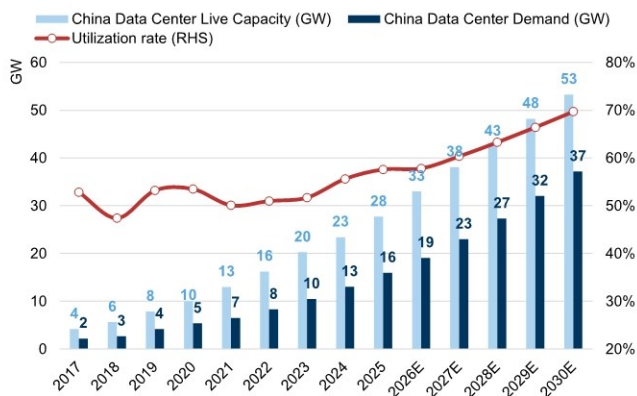
Exhibit 3: Map of China's key data center hubs and 10 clusters



Source: Data compiled by Goldman Sachs Global Investment Research

Exhibit 4: We expect China's data center live capacity to almost double from 28GW in 2025 to 53GW by 2030E

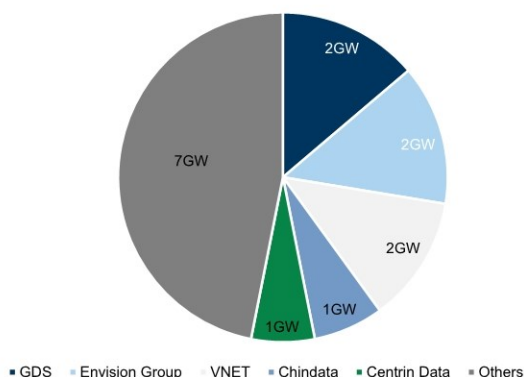
China data center live capacity vs. demand



Source: MIIT, CAICT, Goldman Sachs Global Investment Research

Key players

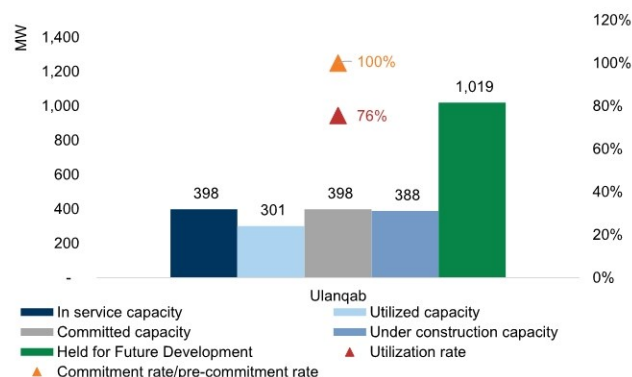
Exhibit 5: We see GDS, VNET and Envision group contributing to the pipeline capacity in Ulanqab
Pipeline capacity share by company



Source: Company data, Sina News, Data compiled by Goldman Sachs Global Investment Research

Exhibit 6: VNET had 398MW in service, 388MW under construction and c.1GW held for future development in Ulanqab as of 1Q26

VNET's data center capacity in Ulanqab



Source: Company data, Data compiled by Goldman Sachs Global Investment Research

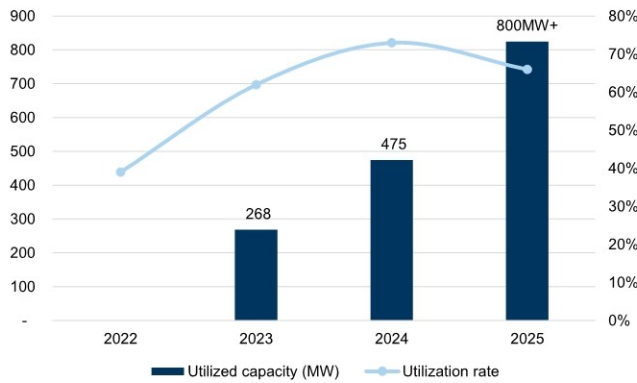
Exhibit 7: Ulanqab is experiencing a surge in large scale data center investments, anchored by investments from data center operators like VNET, GDS, and Centrin Data
Data center projects in Ulanqab

Operator / AI Company	Announced / Signed Date	Investment Amount	Project Scale	Relevant Green Power Projects
Envision Group	Aug-26	NA	Total planned capacity up to 2GW	Company-owned wind farms and green power direct connection; achieving over 80% direct green power supply.
GDS	1H26	Rmb30bn+	Planning 2GW data center campuses	Achieving 80%+ green power coverage via direct green power connection and green power trading
Chindata & Guangdong HEC	Dec-25	NA	Planning for GW-level data center campus	Direct green power connection with "Generation-Grid-Load-Storage" integration
Centrin Data	2024-2025	NA	c.900MW planned capacity with multiple phases of projects	Direct green power connection with "Generation-Grid-Load-Storage" integration; equipped with wind and solar power direct line supply and energy storage systems
VNET	2023-2026	NA	398MW data center capacity in service, 388MW under construction and c.1GW held for future development in Ulanqab as of 1Q26	Green energy projects cooperation with Shandong Hi-Speed Group; 300,000 kW project in Ulanqab involving a Rmb4.4bn investment
Ucloud	Since 2018	NA	5 data center buildings with 140k sqm and 12k+ cabinets; Key customers include Zhipu and vivo	Powered via direct green power connection
Alibaba	Since 2018	NA	Expanded in 2026 with Rmb2.4bn investments	PPA agreement with wind power project
Kuaishou	Jun-20	Rmb10bn+	Self-built data center; 461 acres; supports 300,000 high-performance servers	PUE at 1.2; Green power accounts for 93%;

Source: Company data, Sina News, Data compiled by Goldman Sachs Global Investment Research

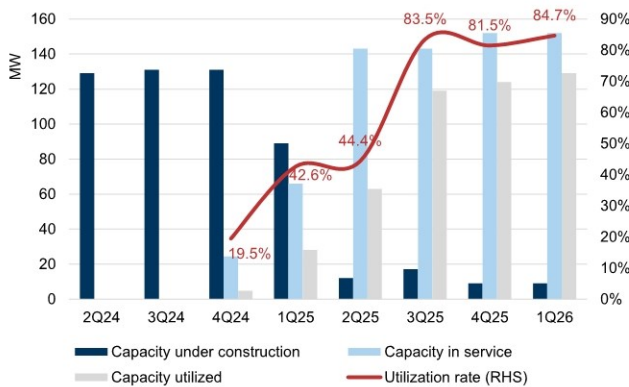
Live data center demand

Exhibit 8: Utilized capacity in Ulanqab grew over 70% yoy to above 800MW in 2025, yet utilization rates fell to 66% due to rapid live capacity growth
Utilized capacity in Ulanqab (MW)



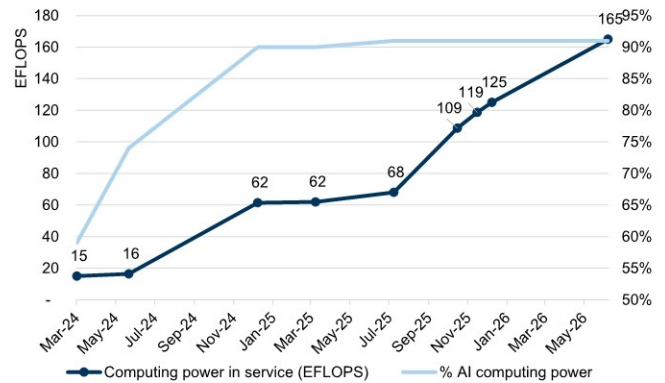
Source: Local government websites, Data compiled by Goldman Sachs Global Investment Research

Exhibit 10: VNET's first Ulanqab campus has reached a high level of utilization
VNET N-OR Campus 01 capacity ramp-up schedule



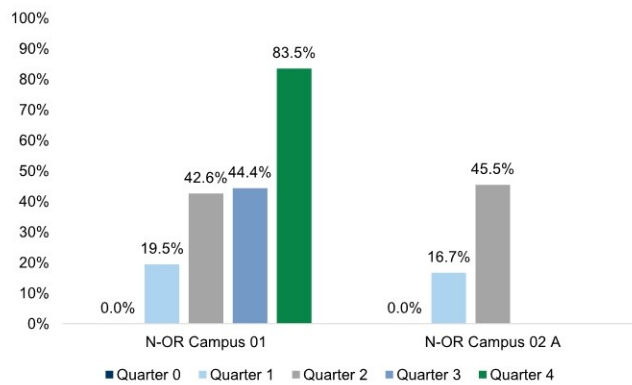
Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 9: Live computing power in service in Ulanqab reached 165 EFLOPS as of Jun 2026
Live computing power in Ulanqab (EFLOPS)



Source: Sina News, Local government websites, Data compiled by Goldman Sachs Global Investment Research

Exhibit 11: VNET: Average utilization rate improved to 40%+/80%+ within 2/4 quarters of capacity delivery
Utilization rate of VNET data center campus in Ulanqab on a cohort basis

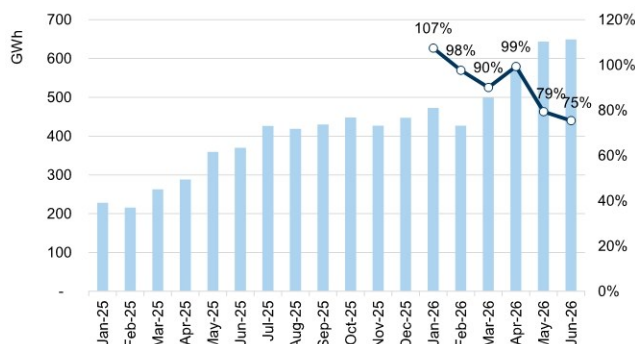


Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Electricity consumption

Exhibit 12: Ulanqab's monthly data center electricity consumption grew +75% yoy to 649GWh in Jun 2026

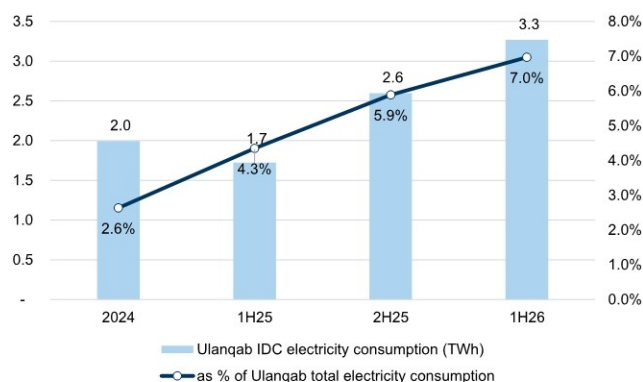
Ulanqab monthly data center electricity consumption (GWh)



Source: Ulanqab NDRC, Data compiled by Goldman Sachs Global Investment Research

Exhibit 13: Electricity consumption of data centers grew 90% yoy to 3.3TWh in 1H26 in Ulanqab, accounting for 7% of total electricity consumption in Ulanqab

Ulanqab data center electricity consumption (TWh) and as % of total electricity consumption



Source: Ulanqab NDRC, Data compiled by Goldman Sachs Global Investment Research

Green power

- **Why Ulanqab for data centers?** i) **Low temperature:** Located at 42°N latitude, Ulanqab has an average annual temperature of only 4.3°C, allowing data centers to use natural cooling to save energy; ii) **Proximity to Beijing:** The city is only 240km away from Beijing; with two direct fiber cables, the latency to Beijing is only 4.2ms; and iii) **Lower electricity prices and rich green power resources:** The electricity price for end data center users ranges around Rmb0.33-0.36 per kWh according to CAICT (vs. Rmb0.6-0.7 per kWh for Beijing or Rmb0.37-0.42 per kWh for the Zhangjiakou cluster near Beijing).
- **Green power supply:** Leveraging rich renewable resources in wind and solar power, Ulanqab has built a sustainable energy grid to support its massive computing clusters. Total renewable energy capacity (including wind and solar power) surpassed **20GW** and is expected to reach 25GW in 2026, with green power accounting for 67% of the city's total electricity mix. Ulanqab has approved 4 integrated "Source-Grid-Load-Storage" projects, which offer green power capacity of 1.6GW and will generate 4.5TWh annually once fully completed. For example, the Ulanqab Wind Power Base Phase I Project, constructed by State Power Investment in Inner Mongolia, is the world's largest single onshore wind project with 6GW capacity; the first 1.2GW phase was successfully grid-connected and accepted in 2025, while the second 1.2GW phase is under construction.
- **Green power's impact on electricity prices:** While green power is not necessarily cheaper than regular grid electricity, data centers leverage green power to meet policy compliance requirements: i) National hubs and new data centers must use green power, with some large scale AI clusters requiring over **80%** green power coverage through trading, certificates, or direct sourcing; and ii) PUE of newly-built data centers should be kept **<1.25**, or **<1.2** for data centers in national hubs and clusters. The two primary ways to obtain green power include: i) direct green power

supply; and ii) green power trading. We see the following factors weighing on the economics of green electricity for data centers: i) generation/PPA prices; ii) green-certificate premium; iii) grid delivery & system charges; iv) dedicated-line and storage capex (direct connection only); and v) any extra backup power needed.

Exhibit 14: Direct Green Power Supply and Green Power Trading & Certificates are two major modes of data center green power consumption

Direct Green Power Supply vs. Green Power Trading & Certificates



Source: CDCC, Data compiled by Goldman Sachs Global Investment Research

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